



AMAN KUMAR

Roll No.:2411EE27

MTech

Electrical Engineering

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Github

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech. (CSSP)	Indian Institute of Technology, Patna	7.6(Current)	2024-Present
B.Tech. (ECE)	MIT World Peace University, Pune	9.48	2017-2021

EXPERIENCE

- Tata Consultancy Services** Feb. 2022 - July 2023
Assistant System Engineer Trainee Kolkata
 - Optimized SQL-driven data pipelines and reporting workflows to improve analytics availability and operational data reliability.
 - Developed Java-based automation solutions to streamline reporting pipelines, reducing manual effort by 20% and improving operational efficiency.
 - Performed production system diagnostics and data-driven root cause analysis to improve performance, stability, and system reliability.
 - Collaborated in Agile teams to support maintenance and delivery of large-scale production systems.
 - Worked with structured datasets and system logs to support analytical workflows and troubleshooting.
- InfoVision Labs India Pvt. Ltd.** Sept. 2021
Software Engineer L1, FTE Pune, India
 - Wrote test cases and debugged Python modules to improve code quality and reliability.
 - Gained practical exposure to software testing lifecycle (STLC).

PROJECTS

- Hybrid Neuro-Symbolic Vehicle Anomaly Detection Platform** Dec 2025 - Present
(Project Demo) IIT Patna
 - Developed a vehicle behavioral threat classification system (Normal / Distracted / Aggressive) from aerial footage using fused ViT visual features and physics-based kinematic divergence on UIT-ADrone and Drone Anomaly datasets.
 - Engineered a retrieval-augmented anomaly intelligence pipeline using FAISS VectorDB and SQLite for contextual memory retrieval, incident search, and scalable video analytics.
 - Developed a real-time tactical analytics dashboard with live object tracking, anomaly visualization, and event-driven threat monitoring for aerial surveillance.
 - Designed a hybrid visual-physics anomaly reasoning pipeline integrating spectral analysis, trajectory dynamics, and Extended Kalman Filter-based motion modeling for interpretable threat assessment.
 - Developed agentic AI inference workflows integrating Gemini APIs and multimodal reasoning for contextual anomaly interpretation and tactical response generation.
- Multi-Agent AI Platform for KPI Intelligence and Decision Support** Dec 2025 - Present
(Github) IIT Patna
 - Architected a self-evaluating **multi-agent AI platform** using Gemini-based LLM orchestration to transform raw marketing data into KPI analysis, forecasting, strategic planning, risk assessment, and executive summaries.
 - Architected and deployed a cloud-hosted **FastAPI backend** integrating Gemini workflows, REST APIs, and production inference pipelines for automated business intelligence and strategic recommendation generation [**Live Deployment**].
 - Engineered **robust multi-agent inference pipelines** integrating schema validation, retry recovery, vector retrieval, prompt engineering, and iterative reasoning refinement for scalable analytics and decision-support systems.
 - Designed **backend architecture** with multi-tier caching, API orchestration, and cloud deployment workflows to improve inference responsiveness and system reliability.
 - Implemented **CI/CD pipelines** using GitHub Actions with automated testing and deployment workflows for reliable production releases.
- AeroViT: Transformer-CNN Framework for Aerial Anomaly Detection** July 2025 - Dec 2025
IIT Patna
 - Developed a hybrid Transformer-CNN anomaly detection framework achieving **89% AUC** on Drone Anomaly and UIT-ADrone datasets with approximately **58× computational efficiency improvement**.
 - Designed hybrid perceptual-loss pipelines combining reconstruction consistency, SSIM, and feature-level anomaly localization for robust aerial video prediction.
 - Implemented FFT-based temporal analysis, motion-consistency evaluation, and embedding-space diagnostics for interpretable anomaly reasoning and model evaluation.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, Java, HTML/CSS
- **AI/ML & Computer Vision:** PyTorch, TensorFlow, Scikit-learn, OpenCV, Ultralytics YOLO, Vision Transformers (ViT)
- **LLM & AI Engineering:** RAG Pipelines, LLM Systems, Prompt Engineering, Multi-Agent AI, Agentic Workflows, Structured Outputs
- **Data & Vector Systems:** FAISS VectorDB, SQLite, Pandas, NumPy, Statistical Analysis, Data Visualization
- **Cloud, Backend & DevOps:** FastAPI, REST APIs, Git, GitHub Actions, CI/CD Pipelines, Render Deployment, Supabase, API Development, Jupyter Notebook, Google Colab, kaggle
- **Software Engineering & Domains:** Backend Development, System Design, API Integration, Async Workflows, Production AI Deployment, AI Systems Deployment, Computer Vision, Real-Time Analytics, Anomaly Detection

ACHIEVEMENTS

- **Qualified GATE 2024**